Maintenance Module

USER MANUAL

Version: 1.0

Last Updated: 27 January 2026

Application: FibreFlow (fibreflow.app)

Module: Maintenance

Classification: Internal Use

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FibreFlow Maintenance Module

Table of Contents

1. Overview

The **Maintenance Module** is FibreFlow's comprehensive fiber network maintenance management system. It handles:

- **Ticket lifecycle management** - From creation to closure
- **Team coordination** - Assign technicians and contractors
- **QA verification** - 12-step verification checklists

- **Handover workflows** - Controlled transitions between teams
- **QContact CRM integration** - Bidirectional sync with FiberTime QContact
- **SLA tracking** - Priority-based response time monitoring
- **Repeat fault escalation** - Automatic detection of recurring issues

Who Uses This Module?

Role	Access Level
Super Admin	Full access to all sections
Manager	Dashboard, tickets, teams, sync, escalations
Technician	View/update assigned tickets, verification steps
QA Reviewer	QA approval, risk acceptance
Coordinator	Dashboard, ticket creation, team assignment

2. Getting Started

Navigating to Maintenance

1. Log in to FibreFlow at **dev.fibreflow.app**
2. In the left sidebar, click **Maintenance** under "Project Management"
3. You will see the Maintenance module with 7 tabs:
    - **Dashboard** - Overview statistics
    - **Work Orders** - Kanban/list view of all tickets
    - **Teams** - Manage maintenance teams
    - **Data Sync** - QContact sync and imports
    - **Escalations** - Repeat fault detection
    - **Handover** - Team transition management
    - **Risk Acceptance** - Conditional QA approvals

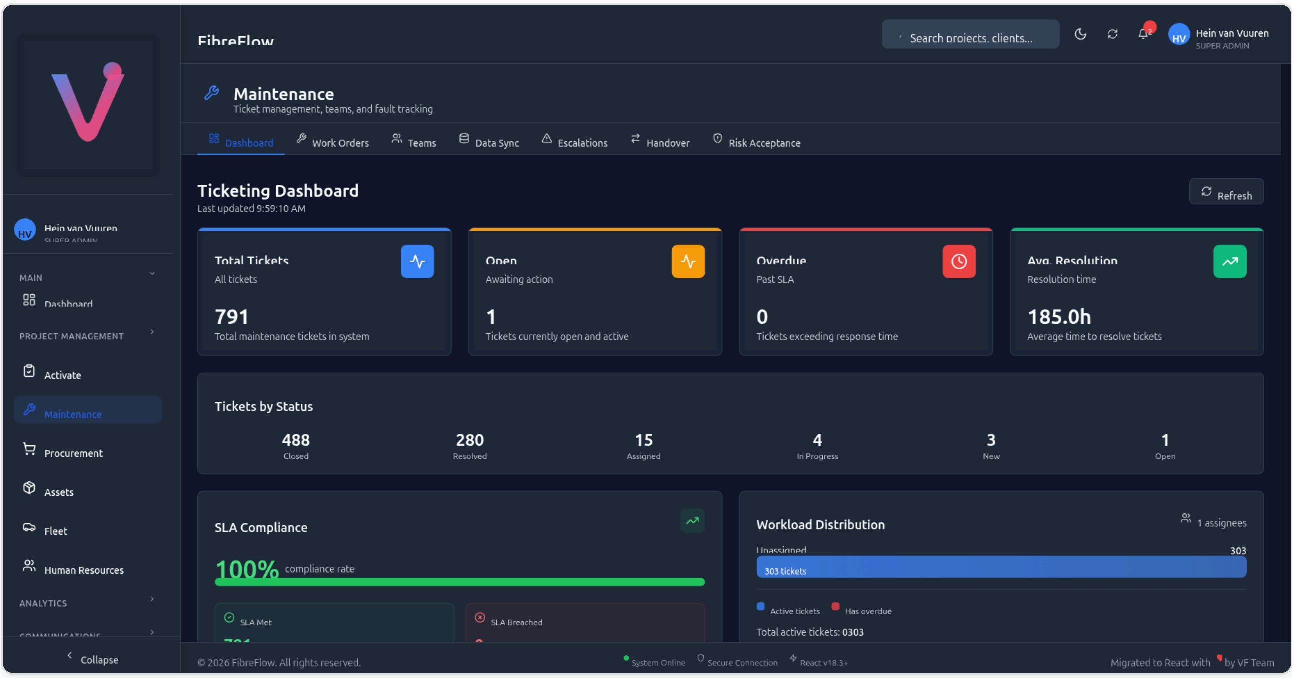

Figure 1: Maintenance Dashboard showing ticket statistics, SLA compliance, and workload distribution

3. Dashboard

The Dashboard provides a real-time overview of your maintenance operations.

Statistics Cards

Card	Description
Total Tickets	All maintenance tickets in the system
Open	Tickets currently open and awaiting action
Overdue	Tickets that have exceeded their SLA response time
Avg. Resolution	Average time to resolve tickets (in hours)

Tickets by Status

A breakdown showing how many tickets are in each status:

- **Closed** - Completed tickets
- **Resolved** - Work completed, pending final closure
- **Assigned** - Assigned to a technician
- **In Progress** - Actively being worked on
- **New** - Newly created, not yet assigned
- **Open** - Awaiting action

SLA Compliance

Shows the percentage of tickets meeting their SLA targets. Each priority level has a different SLA:

Priority	SLA Target
Critical	6 hours
Urgent	12 hours
High	1 day
Normal	3 days
Low	7 days

Workload Distribution

Shows how many tickets are assigned to each team member, helping managers balance workloads.

> ***Tip:*** *The dashboard auto-refreshes every 30 seconds. Click the* ***Refresh*** *button for an immediate update.*

4. Work Orders

The Work Orders tab is the main ticket management view. It provides two ways to view tickets:

1. **Kanban Board** (default) - Visual drag-and-drop columns

2. **Table View** - Traditional filterable list

Switching Between Views

- Click the **Active** toggle to see active (non-closed) tickets
- Click the **Completed** toggle to see resolved/closed tickets

Search

Use the search bar at the top to find tickets by:

- Ticket ID (e.g., FT695563)
- DR number (e.g., DR1732203)
- Description keywords

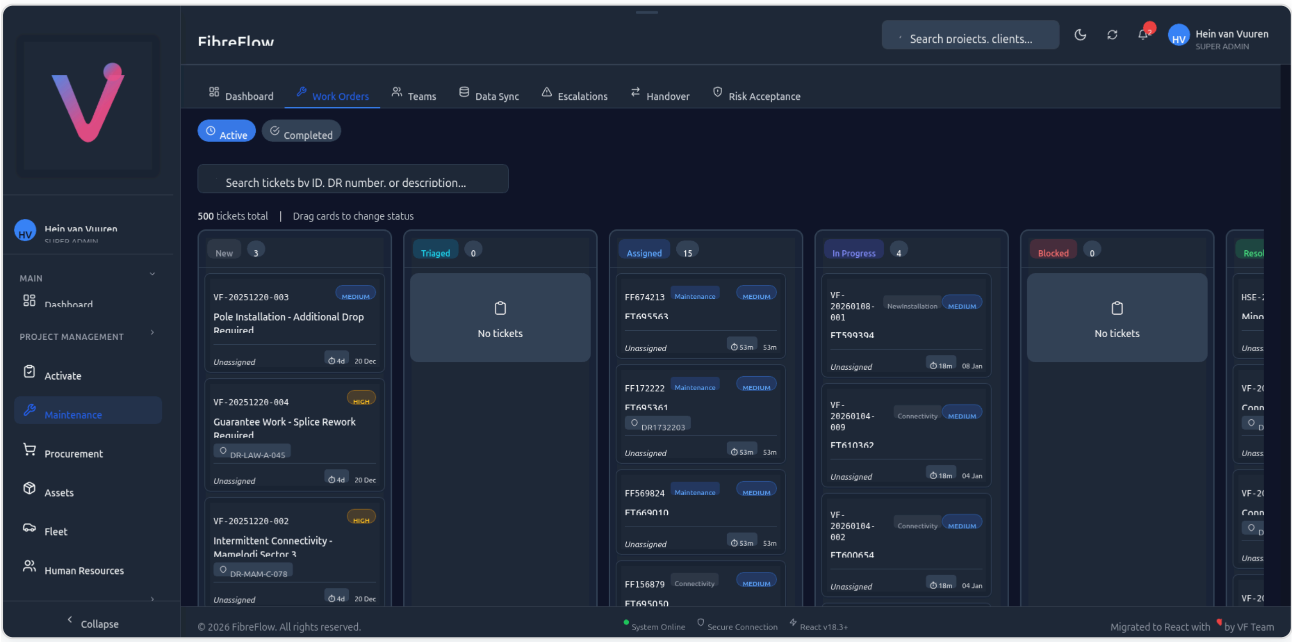

Figure 2: Kanban Board showing tickets organized by status columns with drag-and-drop

5. Kanban Board

The Kanban Board displays tickets as cards organized in status columns. Drag cards between columns to change their status.

Columns

Column	Status	Description
New	`new`	Newly created tickets
Triaged	`triaged`	Assessed and categorized
Assigned	`assigned`	Assigned to a technician
In Progress	`in_progress`	Work is actively underway
Blocked	`blocked`	Work is blocked by an issue
Resolved	`resolved`	Work completed
Closed	`closed`	Ticket fully closed

Card Information

Each card shows:

- **Ticket ID** (e.g., VF-20260108-001)
- **FT Number** (QContact reference, e.g., FT695563)
- **Category badge** (Maintenance, Connectivity, NewInstallation)
- **Priority badge** (Medium, High, Critical)
- **DR Number** (if linked to a drop)
- **Assignee** name
- **SLA indicator** (time remaining)
- **Created date**

Drag-and-Drop

To change a ticket's status:

1. Click and hold a ticket card
2. Drag it to the target column
3. Release to drop
4. The status updates automatically and syncs to QContact

Sorting

Cards within each column are sorted by:

1. **Priority** (Critical first, then Urgent, High, Normal, Low)
2. **Date** (newest first within same priority)

6. Creating a Ticket

Steps to Create a New Ticket

1. Navigate to **Work Orders** tab
2. Click the **+ Create Ticket** button (top-right)
3. Fill in the required fields:

REQUIRED FIELDS

Field	Description
Title	Short description of the issue
Type	Maintenance, New Installation, Modification, ONT Swap, or Incident
Priority	Low, Normal, High, Urgent, or Critical
Source	Where the ticket originated (Manual, QContact, Weekly Report, etc.)

OPTIONAL FIELDS

Field	Description
Description	Detailed description of the issue
DR Number	Drop reference number (auto-populates location data)
Client Name	Customer name
Client Phone	Customer contact number
ONT Serial	ONT device serial number
Assign to Team	Which team should handle this
Assign to User	Specific technician

DR NUMBER LOOKUP

When you enter a DR number, the system automatically looks up:

- Pole number
- PON number
- Zone number
- Project name
- Address
- Coordinates
- Municipality

> ***Tip:*** *You can pre-populate the create form via URL parameters:* `/maintenance/tickets/new?dr_number=DR1234&source=manual&priority=urgent`

Ticket Types

Type	Description	DR Required?
Maintenance	Repair existing fiber network issue	Yes
New Installation	New fiber drop installation	Yes
Modification	Modify existing installation (relocation, upgrade)	Yes
ONT Swap	Replace or swap ONT device	Yes
Incident	Network incident affecting multiple customers	No

Ticket Sources

Source	Description	Created By
Manual	Created via FibreFlow UI	User
QContact	Synced from QContact CRM	Automatic
Weekly Report	Imported from Excel reports	Import wizard
Construction	Created during construction	User
Ad Hoc	Special request	User
Incident	Network incident	User
ONT Swap	ONT replacement request	User

7. Ticket Detail & Lifecycle

Ticket Statuses

Tickets follow a defined workflow. Each status represents a stage in the ticket lifecycle:

```
NEW → ASSIGNED → IN_PROGRESS → PENDING_QA → QA_IN_PROGRESS
                                                        ↓
                                       QA_APPROVED → PENDING_HANDOVER
→ HANDED_TO_OPS → CLOSED
                                       QA_REJECTED → back to
IN_PROGRESS
```

Status	Meaning	Next Steps
Open	Created but not assigned	Assign to technician
Assigned	Assigned, work not started	Start work
In Progress	Work actively underway	Complete verification, submit for QA
Pending QA	Waiting for QA review	QA reviewer picks up
QA In Progress	QA actively reviewing	Approve or reject
QA Rejected	Failed QA review	Technician reworks
QA Approved	Passed QA review	Proceed to handover
Pending Handover	Waiting for team transition	Complete handover
Handed to Ops	Transitioned to operations	Close ticket
Closed	Ticket complete	Terminal state
Cancelled	Ticket cancelled	Terminal state

Priority Levels

Priority	Color	SLA	When to Use
Low	Gray	7 days	Non-urgent, scheduled work
Normal	Blue	3 days	Standard priority
High	Orange	1 day	Important, prioritize
Urgent	Red	12 hours	Immediate attention needed
Critical	Red (pulsing)	6 hours	Drop everything

Fault Cause Attribution

Every maintenance ticket requires a **Fault Cause** - this determines accountability:

Fault Cause	Contractor Liable?	Examples
Workmanship	YES	Improper splicing, loose connections, incorrect routing
Material Failure	No	Defective ONT, faulty cable, broken connector
Client Damage	No	Customer unplugged ONT, cable cut during renovation
Third Party	No	Municipal road work, excavator damage
Environmental	No	Wind damage, lightning strike, water ingress
Vandalism	No	Cable deliberately cut, equipment stolen
Unknown	No	Cannot determine root cause

> ***Important:*** *Fault cause selection directly affects contractor billing. Choose accurately to prevent unfair attribution.*

8. Teams Management

The Teams tab lets you manage maintenance teams and their members.

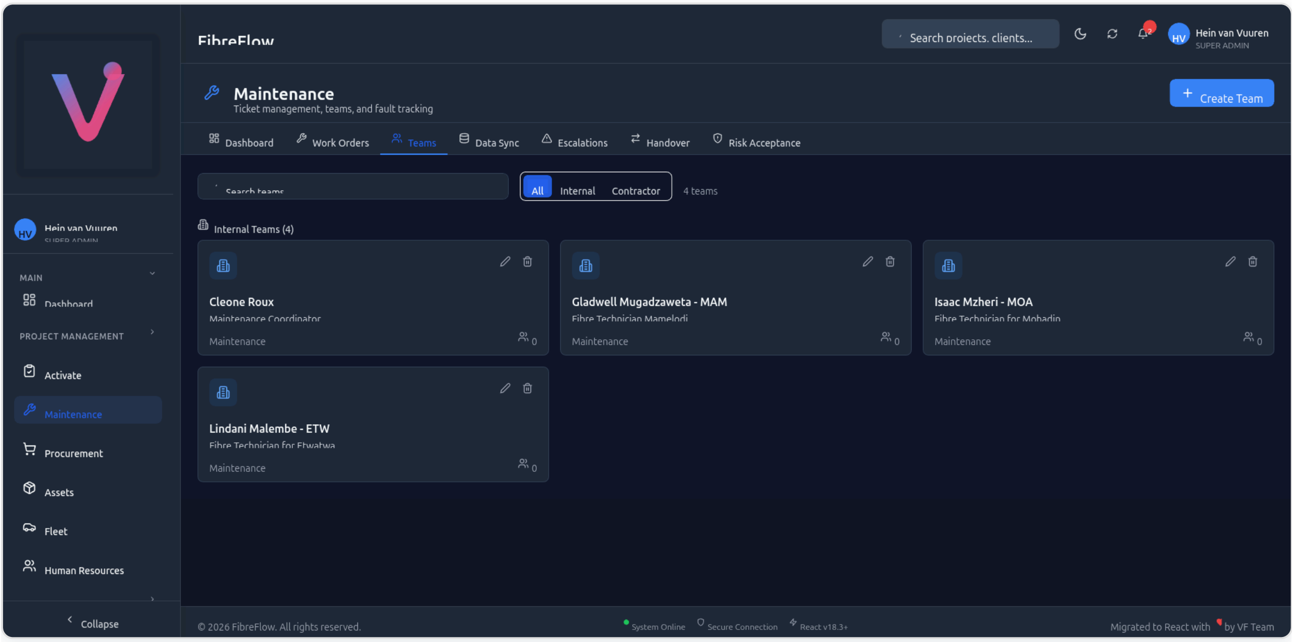

Figure 3: Teams management page showing internal maintenance teams

Features

- **Create Team** - Click "+ Create Team" button (top-right)
- **Search** - Filter teams by name
- **Filter by Type** - All, Internal, or Contractor
- **Edit** - Click the pencil icon on a team card
- **Delete** - Click the trash icon on a team card

Team Types

Type	Description
Internal	In-house maintenance staff
Field	Field technicians
Support	Technical support team
Maintenance	General maintenance team
Installation	New installation team
Contractor	External contractor team

Creating a Team

1. Click **+ Create Team**
2. Enter:
   - **Team Name** (e.g., "Gladwell Mugadzaweta - MAM")
   - **Team Type** (e.g., Maintenance)
   - **Description** (e.g., "Fibre Technician Mamelodi")
3. Click **Save**
4. Add team members after creation

9. Data Sync & QContact Integration

The Data Sync tab manages the bidirectional synchronization between FibreFlow and QContact (FiberTime).

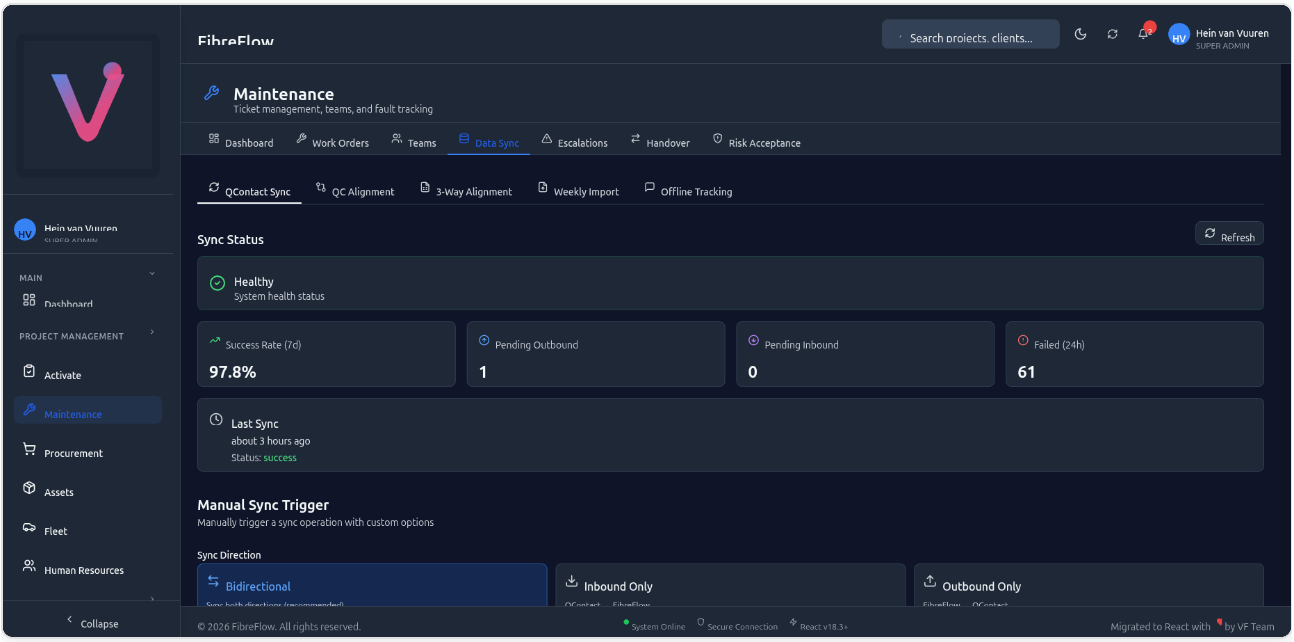

Figure 4: QContact Sync dashboard showing sync health, metrics, and manual trigger

Sub-Tabs

Tab	Purpose
QContact Sync	Main sync dashboard and manual trigger
QC Alignment	Compare QContact and FibreFlow ticket statuses
3-Way Alignment	Cross-reference QC, FF, and Excel data
Weekly Import	Import tickets from Excel weekly reports
Offline Tracking	WhatsApp notification delivery tracking

QContact Sync Dashboard

SYNC STATUS

Metric	Description
Health Status	Green = Healthy, Red = Issues detected
Success Rate (7d)	Percentage of successful syncs in last 7 days
Pending Outbound	FF changes waiting to push to QContact
Pending Inbound	QContact changes waiting to pull into FF
Failed (24h)	Failed sync attempts in last 24 hours
Last Sync	Timestamp and status of most recent sync

MANUAL SYNC TRIGGER

To manually trigger a sync:

1. Choose **Sync Direction**:
    - **Bidirectional** (recommended) - Sync both directions
    - **Inbound Only** - QContact → FibreFlow
    - **Outbound Only** - FibreFlow → QContact
2. Click **Start Sync**
3. Wait for completion (typically 30-60 seconds)
4. Review results in the toast notification

How Sync Works

INBOUND (QCONTACT → FIBREFLOW)

1. FibreFlow queries QContact API for tickets assigned to "Maintenance - Velocity"
2. For each QContact ticket:
    - If not in FibreFlow → **Create** new ticket
    - If already in FibreFlow → **Update** status, category, etc.
3. Status mapping applied (see Section 13)
4. Category/subcategory parsed and stored

OUTBOUND (FIBREFLOW → QCONTACT)

1. When a ticket status changes in FibreFlow
2. If the ticket has a QContact external_id (FT number)
3. FibreFlow pushes the status update to QContact API
4. Reverse status mapping applied

Automatic vs Manual Sync

Type	Trigger	Direction
Automatic (Outbound)	When user changes ticket status in FF	FF → QContact
Manual (Any)	User clicks "Start Sync" button	Configurable
Scheduled (Inbound)	Cron job (if configured)	QContact → FF

10. Escalations & Fault Analysis

The Escalations tab detects and manages repeat fault patterns.

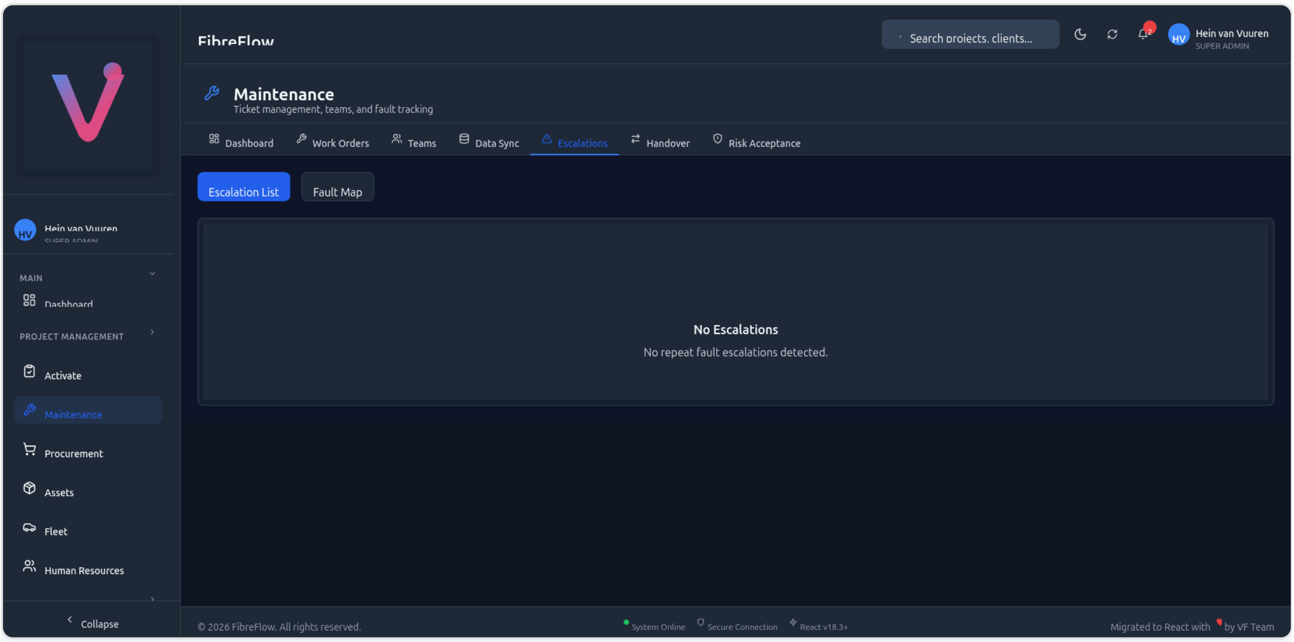

Figure 5: Escalations page showing repeat fault detection status

What Are Escalations?

When the same fault pattern repeats across multiple tickets at the same location, the system automatically creates an **escalation** for infrastructure investigation.

Escalation Scopes

Scope	Description	Example
Pole-Level	Same fault on same pole	3 "Signal loss" tickets on Pole 45
PON-Level	Same fault on same PON	Multiple faults on PON 12
Zone-Level	Same fault across a zone	Pattern of faults in Zone 3
DR-Level	Same fault on same drop	Recurring issue at DR1234

Severity Levels

Severity	Repeat Count	Action Required
Medium	2 repeats	Monitor
High	3-4 repeats	Investigate
Critical	5+ repeats	Immediate investigation

Views

- **Escalation List** - Table of all detected escalations
- **Fault Map** - Geographic visualization of fault patterns

Resolving an Escalation

1. Click on the escalation
2. Review linked tickets
3. Identify root cause
4. Create infrastructure repair ticket

5. Once resolved, mark escalation as resolved

11. Handover Management

The Handover tab manages the controlled transition of tickets between teams.

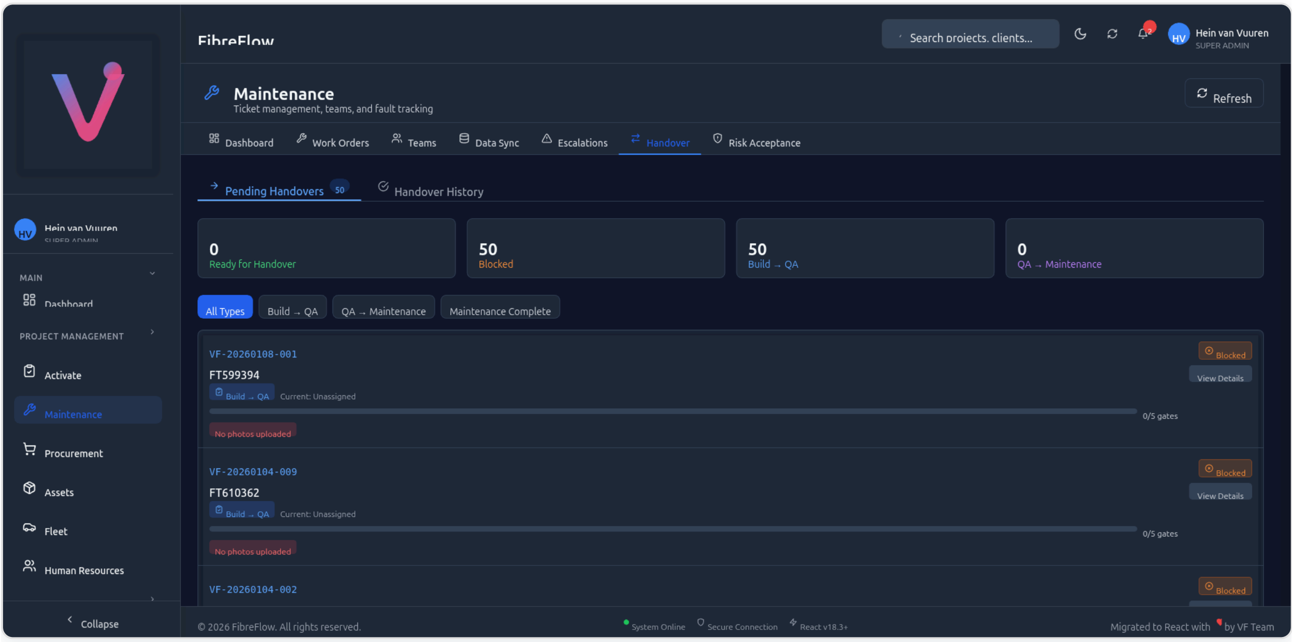

Figure 6: Handover management showing pending handovers with gate progress

Handover Types

Type	From	To
Build → QA	Build/installation team	QA review team
QA → Maintenance	QA team	Operations/maintenance team
Maintenance Complete	Maintenance team	Closed

Handover Gates (5 Mandatory Checks)

Before a handover can proceed, 5 gates must pass:

Gate	Check	Description
1	**Verification Complete**	All 12 verification steps done
2	**QA Status**	QA approval obtained
3	**Documentation**	All paperwork complete
4	**Client Signed Off**	Client has accepted work
5	**Financial**	Payment/billing resolved

The progress bar shows **X/5 gates** passed. If any gate fails, the ticket shows as **Blocked** with specific blocker details.

Sub-Tabs

- **Pending Handovers** - Tickets waiting for handover (with gate status)
- **Handover History** - Completed handover audit trail

Performing a Handover

1. Navigate to **Handover** tab
2. Find the ticket in "Pending Handovers"
3. Click **View Details** to see gate status
4. Resolve any blockers (e.g., upload missing photos)
5. Once all 5 gates pass, click **Proceed with Handover**
6. Add handover notes
7. Confirm

12. Risk Acceptance

The Risk Acceptance tab manages conditional QA approvals where known issues are accepted with conditions.

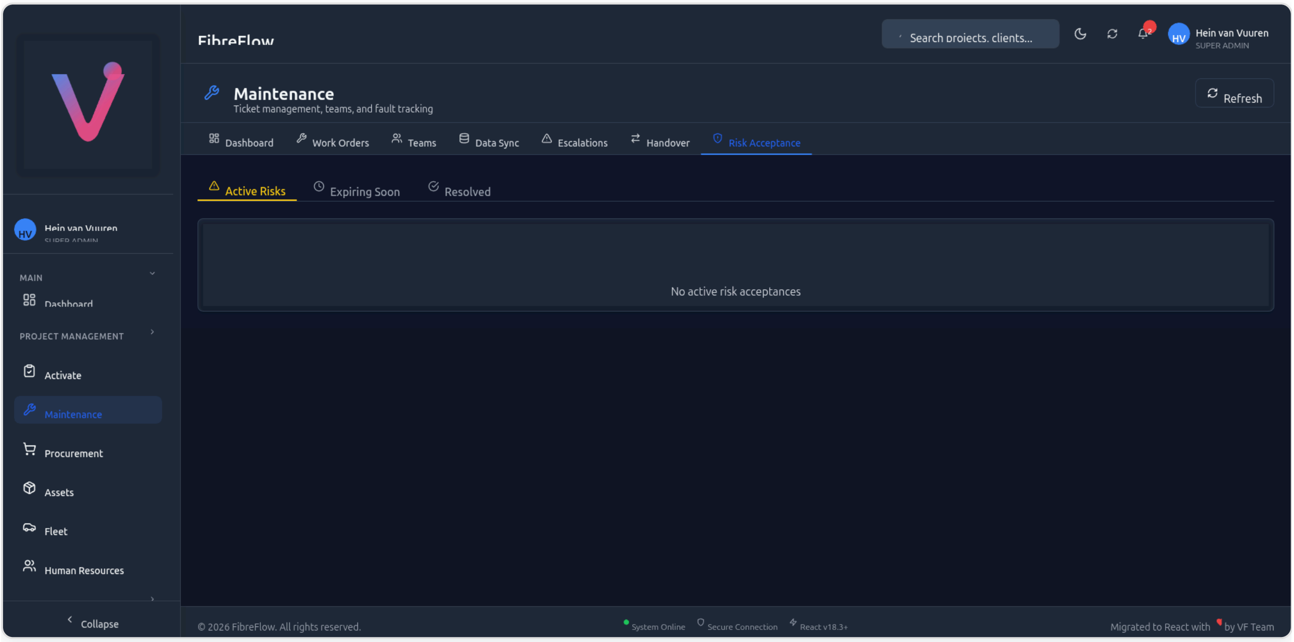

Figure 7: Risk Acceptance page showing active, expiring, and resolved risks

When Is Risk Acceptance Used?

- QA finds an issue that would normally fail approval
- But the project timeline is critical
- QA can **conditionally approve** with documented conditions

Sub-Tabs

Tab	Shows
Active Risks	Currently accepted risks with conditions
Expiring Soon	Risks approaching their expiry date
Resolved	Historical risks that have been resolved

Risk Acceptance Fields

Field	Description
Risk Type	Design, Material, Workmanship, Environmental
Description	What is the risk being accepted
Conditions	What must be true for approval
Accepted By	Who approved the risk
Expiry Date	By when must the issue be resolved
Follow-up Date	When to check on resolution progress

13. QContact vs FibreFlow Comparison

FibreFlow integrates with **QContact** (FiberTime) for maintenance ticket management. Understanding the relationship between the two systems is critical.

System Comparison

Feature	QContact (FiberTime)	FibreFlow
Purpose	CRM - customer contact management	Project management - full workflow
Ticket creation	Manual or via forms	Manual, import, or sync
Status tracking	9 statuses	11 statuses (more granular)
Verification	None	12-step checklist
QA workflow	None	Full QA approval cycle
Handover	None	5-gate handover process
Escalations	None	Automatic repeat fault detection
Risk acceptance	None	Conditional approval workflow
Team management	Basic assignment	Full team management
SLA tracking	Basic	Priority-based with compliance %

Status Mapping (QContact → FibreFlow)

When tickets sync from QContact to FibreFlow, statuses are mapped as follows:

QContact Status	QContact Column	FibreFlow Status	FF Column
New	New	`open`	New
Pending Customer	Pending Customer	`open`	New
Pending Company Response	In Progress	`in_progress`	In Progress
Assigned	Assigned	`assigned`	Assigned
In Progress	In Progress	`in_progress`	In Progress
Escalated	Escalated	`in_progress`	In Progress
In Review	In Review	`pending_qa`	(QA Phase)
Reviewed	In Review	`qa_approved`	(QA Phase)
Solved	Solved	`closed`	Closed
Unsolved - No Response	Unsolved	`cancelled`	Cancelled
Closed	(Historical)	`closed`	Closed

> ***Key Insight:*** *QContact's "Pending Company Response" appears under the "In Progress" column in QContact's UI. FibreFlow maps this to* `in_progress` *to maintain visual alignment between both systems.*

Status Mapping (FibreFlow → QContact)

When changes are made in FibreFlow, they sync back to QContact:

FibreFlow Status	QContact Status
`open`	New
`assigned`	Assigned
`in_progress`	In Progress
`pending_qa`	In Review
`qa_in_progress`	In Review
`qa_rejected`	In Progress
`qa_approved`	Reviewed
`pending_handover`	Reviewed
`handed_to_ops`	Solved
`closed`	Solved
`cancelled`	Unsolved - No Response

Category Mapping

QContact uses categories with separators (`::` or `|`). FibreFlow parses these into category + subcategory:

QContact Category	FF Category	FF Subcategory	FF Ticket Type
`Connectivity::ONT/Gizzu`	Connectivity	ONT/Gizzu	fault
`Connectivity::ONTMove`	Connectivity	ONTMove	ont_swap
`General|Maintenance`	General	Maintenance	fault_repair
`Connectivity::New Installation`	Connectivity	New Installation	installation
`Connectivity::Signal Issue`	Connectivity	Signal Issue	fault
`Connectivity::Link Light`	Connectivity	Link Light	fault
`Connectivity::Bundle`	Connectivity	Bundle	fault

What Syncs Between Systems

Data	Direction	Notes
Ticket status	Bidirectional	Mapped per tables above
Category/subcategory	Inbound only	QC → FF
Ticket type	Inbound only	Derived from category
Contact info	Inbound only	From QContact case details
FT number	Inbound only	Used as external reference
Created date	Inbound only	Preserved from QContact

Side-by-Side Visual Comparison

QContact Kanban (fibertime.qcontact.com):

QContact Kanban *Figure 8: QContact's kanban board showing ticket columns - note "In Progress" column includes "Pending Company Response" tickets*

FibreFlow Kanban (dev.fibreflow.app):

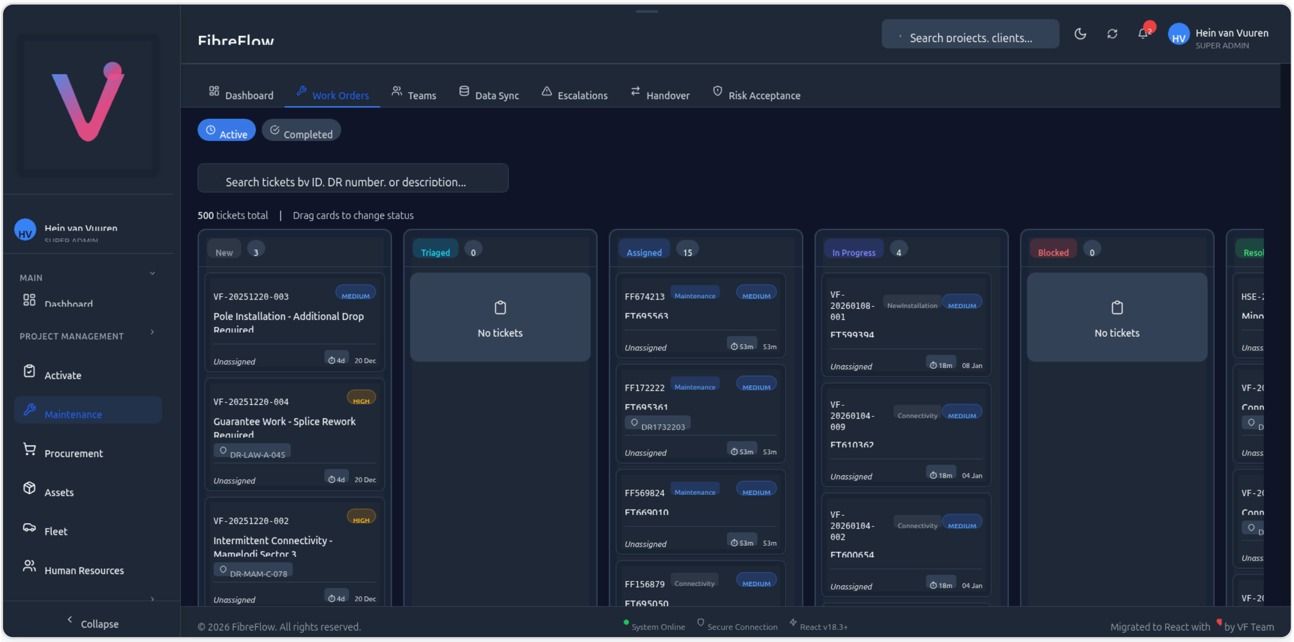

Figure 9: FibreFlow's kanban board - same tickets appear in matching status columns after sync

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14. Common Workflows

Workflow 1: Create and Complete a Maintenance Ticket

1. Go to **Work Orders** → **+ Create Ticket**
2. Fill in: Title, Type (Maintenance), Priority, DR Number
3. Assign to a technician
4. **Technician workflow:**
   - Views assigned ticket in Kanban "Assigned" column
   - Drags to "In Progress" to start work
   - Completes 12 verification steps
   - Uploads photos for each step
   - Selects **Fault Cause** (e.g., "Material Failure - Defective connector")
5. **QA workflow:**
   - Sees "Ready for QA" indicator
   - Reviews verification checklist and photos

  - Approves → ticket moves to "QA Approved"

6. **Handover:**
   - All 5 gates must pass
   - Handover to operations team

7. **Close ticket**

Workflow 2: Handle a QA Rejection

1. Technician submits ticket for QA
2. QA reviewer finds issues → **Rejects**
3. Ticket returns to "In Progress"
4. Technician receives notification
5. Technician fixes the issue, re-uploads photos
6. Resubmits for QA
7. QA approves on second review

Workflow 3: QContact Sync

1. New ticket created in QContact by FiberTime team
2. **Automatic inbound sync** pulls it into FibreFlow
3. Ticket appears in FibreFlow Kanban with correct status
4. FibreFlow user updates the status (e.g., Assigned → In Progress)
5. **Automatic outbound sync** pushes status back to QContact
6. Both systems stay aligned

Workflow 4: Weekly Report Import

1. Go to **Data Sync** → **Weekly Import** tab
2. Click **Upload File** and select the Excel report
3. System parses and validates the data
4. Review preview - fix any errors
5. Click **Confirm Import**
6. Tickets are created and appear in the Kanban board

Workflow 5: Respond to a Repeat Fault Escalation

1. Dashboard shows **Escalation Alert**: "3 signal loss faults on Pole 45"
2. Go to **Escalations** tab
3. Click escalation to see linked tickets
4. Investigate root cause (e.g., damaged patch panel)
5. Create infrastructure ticket for repair
6. Once fixed, mark escalation as resolved

15. Troubleshooting

"Ticket not showing in Kanban"

- Check if you're viewing **Active** or **Completed** toggle
- Verify the ticket status matches a visible Kanban column
- Try refreshing the page
- Check search/filter isn't hiding the ticket

"DR Lookup Failed"

- The DR number may not exist in the SOW (Scope of Work) module
- Check for typos in the DR number
- You can manually fill in location fields if lookup fails

"QContact Sync Failed"

- Check the **Data Sync** → **QContact Sync** tab for error details
- Verify the sync health indicator
- Check **Failed (24h)** counter for recent errors
- Try a manual sync with "Bidirectional" direction
- If persistent, check QContact API availability

"Cannot Move Ticket to Next Status"

- Certain status transitions are restricted (see status flow in Section 7)
- Verify all required fields are filled (fault cause, verification steps)
- Check if you have the right RBAC permissions

"Handover Blocked"

- Check which of the 5 gates are failing
- Common blockers: missing photos, incomplete verification, no fault cause
- Resolve each blocker and the handover will unblock

"SLA Shows Overdue"

- The SLA timer starts when the ticket is created
- Priority determines the SLA window (see Section 3)
- Overdue tickets appear in red on the dashboard
- To clear: resolve the ticket or adjust priority if incorrectly set

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Appendix A: Keyboard Shortcuts

Shortcut	Action
Tab	Navigate between form fields
Escape	Cancel current dialog
Ctrl+Enter	Submit form

Appendix B: RBAC Permissions

Permission Key	Description
`maintenance:tickets:view`	View and manage tickets
`maintenance:tickets:create`	Create new tickets
`maintenance:teams:view`	View and manage teams
`maintenance:data-sync:view`	Trigger syncs, view audit
`maintenance:escalations:view`	View and manage escalations
`maintenance:handover:view`	Manage handover workflow
`maintenance:risks:view`	View and manage risk acceptances

Appendix C: Glossary

Term	Definition
DR	Drop Reference - unique identifier for a fiber drop point
FT Number	FiberTime reference number from QContact
ONT	Optical Network Terminal - customer premises equipment
PON	Passive Optical Network
QContact	FiberTime's CRM system (fibertime.qcontact.com)
SLA	Service Level Agreement - response time target
WIP	Work In Progress
Gizzu	Fiber enclosure/box at drop point

This manual is maintained by the FibreFlow development team. For questions or corrections, contact the system administrator.

Document generated: 27 January 2026